

MODULE 3 _ASSIGNMENT

1. Scenario: Automating file backup and Reporting to the system. Create a shell script called "backup_manager.sh" that performs the following tasks incorporating the concepts suggested.

```
rohit45@rohit45-VirtualBox:~$ cat backup_manager.sh
```

```
#!/bin/bash
```

```
echo
```

```
echo "BACKUP MANAGER"
```

```
echo "_____"
```

```
sdir="$1"
```

```
ddir="$2"
```

```
ft="$3"
```

```
echo "CHECK DETAILS"
```

```
echo "_____"
```

```
echo "SOURCE DIR = $sdir"
```

```
echo "DESTINATION DIR = $ddir"
```

```
echo "FILE TYPE = $ft"
```

```
echo
```

```
declare -A FILE_SIZE
```

```
shopt -s nullglob
```

```
for a in "$sdir"/*. "$ft"
```

```
do
```

```
    fname=$(basename "$a")
```

```
    size=$(stat -c %s "$a")
```

```
    FILE_SIZE["$fname"]="$size"
```

```
done
```

```
shopt -u nullglob
```

```
echo
```

```
if [ "${#FILE_SIZE[@]}" -eq 0 ]; then

    echo "NO FILE WITH THE GIVEN EXTENSION .$ft IS FOUND. Exiting..."

    exit

else

    echo "SOURCE_DIR CHECK SUCCEEDED"

fi

export BACKUP_COUNT="${#FILE_SIZE[@]}"

echo "NUMBER OF FILES CAPTURED IN THE GIVEN FORMAT : $BACKUP_COUNT"

echo

echo "FILE REPORT"

echo "_____ "

for fname in "${!FILE_SIZE[@]}"
do
    echo "$fname (SIZE : ${FILE_SIZE[$fname]} BYTES)"
done

echo

if [ -d "$ddir" ]; then

    echo "BACKUP_DIRECTORY EXISTS "$ddir""

else

    echo "BACKUP_DIRECTORY NOT FOUND, CREATING $ddir"

    mkdir -p "$ddir"

    if [ $? -ne 0 ]; then

        echo "BACKUP_DIRECTORY CREATION FAILED. Exiting..."

        exit 1

    fi

fi

echo

backup_files=0
total_size=0
backup_files=0
total_size=0
```

```
for filename in "${!FILE_SIZE[@]}"
do
    sourcev="$sdir/$filename"
    backupv="$ddir/$filename"

    if [ ! -e "$backupv" ]; then
        cp "$sourcev" "$backupv"
        if [ $? -eq 0 ]; then
            ((backup_files++))
            ((total_size += FILE_SIZE["$filename"]))
            echo "NEW FILE CREATED AND BACKUP SUCCESS"
        fi
    elif [ "$sourcev" -nt "$backupv" ]; then
        cp "$sourcev" "$backupv"
        if [ $? -eq 0 ]; then
            ((backup_files++))
            ((total_size += FILE_SIZE["$filename"]))
            echo "NEW CHANGES FOUND..."
            echo "BACKUP SUCCESS OVERWRITTEN ON OLD BACKUP FILE"
        fi
    else
        echo "FILE $filename IS UP-TO-DATE"
    fi
done

echo
echo "PATH TO THE BACKUP DIRECTORY : $ddir"
echo "OUTPUT REPORT PROCESSING ON $ddir/backup_report.log"

echo "$(date)" >> "$ddir/backup_report.log"
echo "TOTAL FILES PROCESSED : $backup_files" >> "$ddir/backup_report.log"
```

```
echo "TOTAL FILE SIZE PROCESSED : $total_size" >> "$ddir/backup_report.log"
echo "PATH TO THE BACKUP DIRECTORY : $ddir" >> "$ddir/backup_report.log"
```

```
echo "REPORT SAVED, BACKUPS COMPLETED"
```

```
echo
```

```
rohit45@rohit45-VirtualBox:~$ cd sdr
```

```
rohit45@rohit45-VirtualBox:~/sdr$ ls
```

```
filec1.csv filec2.csv filec3.csv files1.sh files2.sh files3.sh files4.sh file1.txt file2.txt file3.txt file4.txt newfile.txt
```

```
rohit45@rohit45-VirtualBox:~/sdr$ cd ddr
```

```
bash: cd: ddr: No such file or directory
```

```
rohit45@rohit45-VirtualBox:~/sdr$ ./backup_manager.sh "sdr" "ddr" "txt"
```

```
bash: ./backup_manager.sh: No such file or directory
```

```
rohit45@rohit45-VirtualBox:~/sdr$ cd ..
```

```
rohit45@rohit45-VirtualBox:~$ ./backup_manager.sh "sdr" "ddr" "txt"
```

```
BACKUP MANAGER
```

```
CHECK DETAILS
```

```
SOURCE DIR = sdr
```

```
DESTINATION DIR = ddr
```

```
FILE TYPE = txt
```

```
SOURCE_DIR CHECK SUCCEEDED
```

```
NUMBER OF FILES CAPTURED IN THE GIVEN FORMAT : 5
```

```
FILE REPORT
```

```
file1.txt (SIZE : 830 BYTES)
```

```
file2.txt (SIZE : 896 BYTES)
```

```
newfile.txt (SIZE : 40 BYTES)
```

```
file3.txt (SIZE : 706 BYTES)
```

```
file4.txt (SIZE : 63 BYTES)
```

BACKUP_DIRECTORY NOT FOUND, CREATING ddr

NEW FILE CREATED AND BACKUP SUCCESS

NEW FILE CREATED AND BACKUP SUCCESS

NEW FILE CREATED AND BACKUP SUCCESS

NEW FILE CREATED AND BACKUP SUCCESS

NEW FILE CREATED AND BACKUP SUCCESS

PATH TO THE BACKUP DIRECTORY : ddr

OUTPUT REPORT PROCESSING ON ddr/backup_report.log

REPORT SAVED, BACKUPS COMPLETED

```
rohit45@rohit45-VirtualBox:~$ cd ddr
```

```
rohit45@rohit45-VirtualBox:~/ddr$ ls
```

```
backup_report.log  file1.txt  file2.txt  file3.txt  file4.txt  newfile.txt
```

```
rohit45@rohit45-VirtualBox:~/ddr$ cat backup_report.log
```

```
Mon Feb  2 08:11:17 PM IST 2026
```

TOTAL FILES PROCESSED : 5

TOTAL FILE SIZE PROCESSED : 2535

PATH TO THE BACKUP DIRECTORY : ddr

```
rohit45@rohit45-VirtualBox:~/ddr$ cd ..
```

```
rohit45@rohit45-VirtualBox:~$ ./backup_manager.sh "sdr" "ddr" "txt"
```

BACKUP MANAGER

CHECK DETAILS

SOURCE DIR = sdr

DESTINATION DIR = ddr

FILE TYPE = txt

SOURCE_DIR CHECK SUCCEEDED

NUMBER OF FILES CAPTURED IN THE GIVEN FORMAT : 5

FILE REPORT

file1.txt (SIZE : 830 BYTES)

file2.txt (SIZE : 896 BYTES)

newfile.txt (SIZE : 40 BYTES)

file3.txt (SIZE : 706 BYTES)

file4.txt (SIZE : 63 BYTES)

BACKUP_DIRECTORY EXISTS ddr

FILE file1.txt IS UP-TO-DATE

FILE file2.txt IS UP-TO-DATE

FILE newfile.txt IS UP-TO-DATE

FILE file3.txt IS UP-TO-DATE

FILE file4.txt IS UP-TO-DATE

PATH TO THE BACKUP DIRECTORY : ddr

OUTPUT REPORT PROCESSING ON ddr/backup_report.log

REPORT SAVED, BACKUPS COMPLETED

rohit45@rohit45-VirtualBox:~\$ cd ddr

rohit45@rohit45-VirtualBox:~/ddr\$ cat backup_report.log

Mon Feb 2 08:11:17 PM IST 2026

TOTAL FILES PROCESSED : 5

TOTAL FILE SIZE PROCESSED : 2535

PATH TO THE BACKUP DIRECTORY : ddr

Mon Feb 2 08:11:44 PM IST 2026

TOTAL FILES PROCESSED : 0

TOTAL FILE SIZE PROCESSED : 0

PATH TO THE BACKUP DIRECTORY : ddr

rohit45@rohit45-VirtualBox:~/ddr\$